

Day 3:- Combinational Assignment

- i) Implement the following functions using
i) 2:1 mux and NOT gates ii) 4:1 mux

a) $F(A, B, C) = \overline{A}C + B\overline{C}$

choose C as select line.

For 2:1 mux

$$F = S I_0 + \overline{S} I_1$$

when $C=0$, $F=B$

when $C=1$, $F=\overline{A}$

Hence $S=C$, $I_0=B$, $I_1=\overline{A}$

$$F = \text{MUX}(C, B, \overline{A})$$

- b) Using 4:1 mux

Take A, B as select lines

A	B	F
0	0	0
0	1	1
1	0	0
1	1	0

$$S_1 = A, S_0 = B$$

$$I_0 = 0, I_1 = 1, I_2 = 0, I_3 = 0$$

$$F = \text{MUX}(A, B, C, 1, 0, 0, 0)$$

- 2) Implement the functionality of a full adder using 2:1 mux only

$$S = A \oplus B \oplus \text{cin}$$

$$\text{Cout} = AB + B\text{cin} + A\text{cin}$$

Using 2:1 mux with select A

$$A \oplus B = \overline{A}B + A\overline{B}$$

$$S_{\text{select}} = A$$

$$I_0 = B$$

$$I_1 = \overline{B}$$

$$X = A \oplus B$$

$$S = (A \oplus B) \oplus \text{cin}$$

$$X = A \oplus B$$

$$S = X \oplus \text{cin}$$

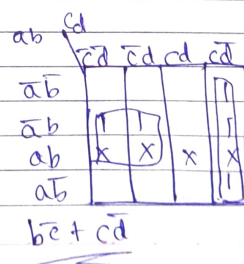
using 2:1 mux

X → select line

$$S = X(\text{cin}) + \overline{X}(\overline{\text{cin}})$$

$$S = X$$

- 3) Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + \sum d(7, 13, 14, 15)$ and implement the SOP expression using 4:1 mux



$$f(a, b, c, d) = b\overline{c} + \overline{a}d + cd$$

$$I_0 = \text{cin}, I_1 = \overline{\text{cin}}$$

$$S = (A \oplus B) \oplus \text{cin}$$

Carry:

$$\text{Cout} = AB + A\text{cin} + B\text{cin}$$

$$F = S I_0 + \overline{S} I_1$$

$$X = A \oplus B$$

$$X = 0$$

$$A = B$$

$$A \quad B$$

$$0 \quad 0$$

$$1 \quad 1$$

when $A=B=0$

$$\text{Cout} = 0$$

when $A=B=1$, $\text{Cout} = 1$

whenever $X=0$

$$\text{Cout} = A$$

$$X = 1$$

$$A \neq B$$

$$A \quad B$$

$$0 \quad 1$$

$$1 \quad 0$$

when $A=0, B=1$

$$\text{Cout} = 0 + (0)\text{cin} + 1 \cdot \text{cin} = \text{cin}$$

when $A=1, B=0$

$$\text{Cout} = 0 + 1\text{cin} + 0 = \text{cin}$$

$$\text{Cout} = \text{cin}$$

$$Y = S I_0 + \overline{S} I_1$$

$$S = X = A \oplus B$$

$$X = 0 \Rightarrow A$$

$$X = 1 \Rightarrow \text{cin}$$

$$I_0 = A, \text{Cout} = XA + X\text{cin}$$

$$I_1 = \text{cin}, \text{Cout} = (A \oplus B)A + (A \oplus B)\text{cin}$$

Implement using 4:1 mux
choose B and C as select lines

$$S_1 = B \quad S_0 = C$$

$$BC = 00, \text{ then } F = 0 \Rightarrow I_0 = 0$$

$$BC = 01, \text{ then } F = D' \Rightarrow I_1 = D'$$

$$BC = 10, \text{ then } F = 1 \Rightarrow I_2 = 1$$

$$BC = 11, \text{ then } F = D' \Rightarrow I_3 = D'$$

$$F = \underline{BC + CD}$$